École Nationale Supérieure d'Informatique

Laboratory of Embedded Systems & IoT

GNS3 Auto-Config DHCP Portal

Network Automation & Programmability Project

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1 Introduction

The GNS3 Auto-Config DHCP Portal is an automated network provisioning solution designed to bridge modern web paradigms with low-level network engineering. The project aims to eliminate manual CLI intervention by translating web-based user inputs into direct, programmatic configurations on emulated Cisco and Linux hardware.

2 System Architecture

The project implements a 3-tier automation architecture:

- **Presentation Layer:** A Next.js 15 frontend providing a dashboard for network parameter entry.
- **Logic Layer:** A Flask-based REST API that performs IP validation and generates configuration strings.
- **Execution Layer:** An automation engine using the Paramiko library to inject CLI commands via SSH into a GNS3 virtual environment.

3 Technical Implementation

3.1 Backend Logic & SSH Pipeline

The backend leverages Python's `ipaddress` module to ensure subnet integrity before establishing a stateful SSH socket connection.

```python
import paramiko

def apply_config(ip, username, password, config_commands):
    ssh = paramiko.SSHClient()
    ssh.set_missing_host_key_policy(paramiko.AutoAddPolicy())
    ssh.connect(ip, username=username, password=password)

    remote_conn = ssh.invoke_shell()
    remote_conn.send("conf t\n")
    for cmd in config_commands:
        remote_conn.send(f"{cmd}\n")
    remote_conn.send("end\n")
```

Listing 1: Configuration Application Logic

3.2 GNS3 Topology Integration

To allow the host machine to communicate with the emulated Cisco Router, a **Cloud Node** was utilized to bridge the virtual lab to the physical network adapter. This enables the Flask server (running on the host) to reach the Router's management interface.

4 Conclusion

This project demonstrates the transition from traditional manual networking to **Software Defined Networking (SDN)**. By decoupling the configuration logic from the hardware, we achieve faster deployment times and reduce human error in IP management.